

CPARMS-WMF: Clustering and Pairwise Association Rule Mining Signal for WMF with Sparse Implicit Feedback

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Abstract

Weighted matrix factorization (WMF) struggles when interactions are sparse and side information is not always available. We investigate whether patterns derived only from the interaction matrix can improve personalized Top- K recommendation for users with short histories.

We propose CPARMS-WMF, a feature-free framework that clusters users and items, mines single-antecedent rules separately from liked and disliked feedback, and adds the resulting user-item scores to WMF as positive and negative targets. We evaluate it on five Amazon datasets using a global temporal split and five random seeds against ItemPop, Standard-WMF, CoFactor, RME, NeuMF, and LightGCN.

Keywords: Recommender systems, Sparse implicit feedback, Weighted matrix factorization, Clustering, Association rule mining

1 Introduction

Recommender systems help people find relevant products and content in large catalogs, but recommendation quality often decreases for users with only one or a few recorded interactions.

Collaborative filtering, especially matrix factorization (MF) [1], remains widely used for its accuracy and scalability; it learns directly from interaction data, without requiring any side information. Weighted matrix factorization (WMF) [2] adapts MF to implicit feedback by assigning different confidence levels to observed and unobserved interactions. Alternating least squares (ALS) can optimize it efficiently. Both MF and WMF still need enough interaction history: when a user’s history contains only a few interactions, the corresponding latent factors cannot be estimated reliably, and Top- K ranking quality drops; highly sparse datasets make this even worse [3].

One common solution to user sparsity is to add side information such as user profiles, demographic attributes, item descriptions, or cross-domain knowledge [4–6]. Side information can improve accuracy, but it is not always available. This gap motivates feature-free methods, which do not require user attributes or item metadata.

The interaction matrix contains patterns beyond the history of any single user or item. Clustering can reveal group-level preferences [7]. Association rule mining can reveal directional relationships, and previous work has combined item clustering with association rule mining to reduce sparsity [8]. Co-occurrence is another source of information. CoFactor adds an item–item co-occurrence term to WMF [9]. RME extends this idea with three co-occurrence terms: items that users often like together, items that users often dislike together, and users who like the same items [10].

These studies use clustering, association rules, or co-occurrence, but none of them combines clustering, association rule mining, and weighted matrix factorization in a single model. Clustering captures shared preference structure, association rules identify directional relationships, and WMF uses this information for personalized ranking. CPARMS-WMF combines these signals within a single objective and, like RME, uses both liked and disliked feedback.

We propose **CPARMS-WMF**, a feature-free framework for personalized recommendation under sparse implicit feedback. Its **Clustering and Pairwise Association Rule Mining Signal (CPARMS)** is derived only from the user–item interaction matrix. CPARMS builds two signals, one from liked feedback and one from disliked feedback. For each signal, it clusters users and items, mines single-antecedent rules from user-cluster, item-cluster, and item-level tokens, and converts those rules into a user–item signal. Two CPARMS terms add these signals to the ALS objective: the liked signal raises predicted scores, and the disliked signal lowers them, each with its own weight.

We evaluate users with at least one interaction in the fit window and at least one positive target in the evaluation window. Users and items that appear only in the evaluation window are excluded from training and the candidate set, and their evaluation events are not scored. Results are reported overall and separately for users with one, two, and three or more interactions in the fit window.

CPARMS-WMF is compared with six feature-free baselines in three groups: a non-personalized popularity baseline (ItemPop); three models that share the WMF

backbone, namely Standard-WMF [2], CoFactor [9], and RME [10]; and two models with different backbones, namely NeuMF [11] and LightGCN [12].

The main contributions are:

1. CPARMS-WMF is introduced as a feature-free method for recommendation under sparse implicit feedback that does not require user attributes or item metadata.
2. CPARMS combines user and item clustering with pairwise association rule mining. It builds separate signals from liked and disliked feedback, and each signal uses user-cluster, item-cluster, and item-level rules from the transactions.
3. CPARMS is integrated into WMF through two additional terms optimized by alternating least squares.
4. CPARMS-WMF is evaluated on five Amazon domains using five random seeds against six baselines. NDCG@10 is the primary metric, with NDCG@20, NDCG@50, NDCG@100, and NDCG@200 reported as secondary measures. The analysis covers mean performance and per-user paired t-tests, both overall and for users with one, two, and three or more interactions.

2 Related Work

We first review the popularity baseline and matrix factorization for implicit feedback. We then review methods that add side information, methods that extract additional signals from the interaction matrix, and neural and graph-based recommenders.

2.1 Non-personalized Recommendation

Ranking candidate items by global popularity, identically for every user, is the standard non-personalized reference point in recommender evaluation [13, 14]. We call this baseline ItemPop. Because it does not use individual user histories, a personalized model that cannot outperform it has not demonstrated an accuracy benefit from personalization under this evaluation protocol. Popularity remains a strong Top- K baseline: prior work shows that a naive TopPop method can rival sophisticated personalized algorithms and that highly popular items can skew Top- K accuracy [15].

2.2 Matrix Factorization and Weighted Matrix Factorization

Matrix factorization (MF) is a core recommendation technique. It places users and items in a shared low-dimensional latent space and estimates preferences from how their latent factors interact [1].

Weighted matrix factorization (WMF) extends MF to implicit-feedback data by modeling binary preference alongside confidence weights. WMF learns its latent factors by minimizing a confidence-weighted WMF loss, most often with alternating least squares [2].

Both methods remain sensitive to sparsity: when observed interactions are limited, estimates become unreliable, and Top- K ranking quality drops [3]. This problem motivates additional signals derived from interactions.

2.3 Hybrid Matrix Factorization with Side Information

A common approach to reducing sparsity is to use side information with collaborative filtering. Content-boosted matrix factorization adds item attributes directly to the factorization, improving predictive accuracy and making recommendations easier to interpret [4].

Cross-domain matrix factorization transfers information from related domains by using item metadata and semantic relationships as links between source and target items. It is especially helpful for new users [5]. More broadly, hybrid recommender systems combine collaborative and content-based filtering so that each approach can address the weaknesses of the other [6].

2.4 Hybrid Matrix Factorization without Side Information

Interaction-derived methods draw additional signals from user-item data without requiring external attributes. Clustering-based approaches identify preference structures shared by users or items [7], while association-rule approaches mine directional item relationships from transaction data. Some studies combine the two techniques or connect one of them to matrix factorization. For example, one approach reduces the dimensionality of the item space through clustering and then applies association rule mining to implicit interactions [8]. Another line of work adds co-occurrence terms to the factorization objective. CoFactor adds an item-item co-occurrence term, supplying information for items with few observed interactions [9]. RME extends CoFactor with a co-occurrence term for disliked items and a co-occurrence term for users [10]. To our knowledge, none of the studies reviewed above integrates clustering, association rule mining, and weighted matrix factorization within one model.

CoFactor is a relevant weighted matrix factorization baseline because it links implicit-feedback WMF with an item-item co-occurrence signal derived from the same interaction matrix [9]. CoFactor augments Standard-WMF with an item-item shifted positive pointwise mutual information (SPPMI) term. Using the notation of this paper, a CoFactor-style objective is

$$\begin{aligned} \mathcal{L}_{\text{CoFactor}} = & \sum_{u \in \mathcal{U}} \sum_{i \in \mathcal{I}} c_{ui} (\pi_{ui} - \mathbf{p}_u^\top \mathbf{q}_i)^2 \\ & + \gamma_{\text{co}} \sum_{(i,j) \in \mathcal{M}} (m_{ij} - \mathbf{q}_i^\top \mathbf{z}_j - b_i - d_j - g)^2 \\ & + \lambda (\|\mathbf{P}\|_F^2 + \|\mathbf{Q}\|_F^2) + \lambda_{\text{ctx}} \|\mathbf{Z}\|_F^2. \end{aligned} \quad (1)$$

Here π_{ui} is the binary implicit preference, c_{ui} is its WMF confidence weight, $\mathbf{p}_u$ and $\mathbf{q}_i$ are user and item factors, m_{ij} is the SPPMI value for item i and context item j , $\mathcal{M}$ indexes nonzero SPPMI entries, $\mathbf{z}_j$ is a context factor, b_i and d_j are item-side and context-side biases, g is a global intercept, γ_{co} controls the co-occurrence-term weight, and λ_{ctx} regularizes the context factors. The shared item factor $\mathbf{q}_i$ connects the WMF term to the SPPMI term. We compute the SPPMI entries from item co-occurrence counts c_{ij} , where c_{ij} is the number of users who positively interacted with both items

i and j , and we set the diagonal to zero:

$$\text{SPPMI}_{ij} = \max \left(\log \frac{c_{ij}N}{f_i f_j} - \log k, 0 \right), \quad (2)$$

where N is the total co-occurrence count over nonzero entries, f_i and f_j are the marginal co-occurrence counts of items i and j , and k is the number of negative samples, whose logarithm shifts the pointwise mutual information. CoFactor therefore adds a signal derived from interactions, but this signal encodes item-level co-occurrence rather than group-level user preference patterns.

Equation (1) keeps the co-occurrence term weighted and the WMF term at unit scale. The original formulation instead applies a relative scale ℓ to the WMF term and its user and item regularizers. Dividing the objective by ℓ gives Eq. (1) with $\gamma_{\text{co}} = 1/\ell$. Our hyperparameter search keeps this relationship: we sample ℓ and set $\gamma_{\text{co}} = 1/\ell$, and we sample λ_{ctx} separately (Section 4.5).

RME follows the same idea but uses three SPPMI terms instead of one [10]. The first is the CoFactor term over items that users like together. The second is a term over items that users dislike together, computed in the same way from disliked feedback. The third is a user-user term over users who like the same items. Using the same notation, the RME objective is

$$\begin{aligned} \mathcal{L}_{\text{RME}} = & \sum_{u \in \mathcal{U}} \sum_{i \in \mathcal{I}} c_{ui} (\pi_{ui} - \mathbf{p}_u^\top \mathbf{q}_i)^2 \\ & + \gamma^+ \sum_{(i,j) \in \mathcal{M}^+} (m_{ij}^+ - \mathbf{q}_i^\top \mathbf{z}_j^+ - b_i^+ - d_j^+ - g^+)^2 \\ & + \gamma^- \sum_{(i,j) \in \mathcal{M}^-} (m_{ij}^- - \mathbf{q}_i^\top \mathbf{z}_j^- - b_i^- - d_j^- - g^-)^2 \\ & + \gamma^U \sum_{(u,v) \in \mathcal{N}} (n_{uv} - \mathbf{p}_u^\top \mathbf{w}_v - e_u - h_v - g^U)^2 \\ & + \lambda (\|\mathbf{P}\|_F^2 + \|\mathbf{Q}\|_F^2) + \lambda_{\text{ctx}} (\|\mathbf{Z}^+\|_F^2 + \|\mathbf{Z}^-\|_F^2 + \|\mathbf{W}\|_F^2). \end{aligned} \quad (3)$$

Here m_{ij}^+ and m_{ij}^- are the SPPMI values of Eq. (2) computed from liked and disliked feedback, n_{uv} is the SPPMI value for users u and v computed from liked feedback, $\mathbf{z}_j^+$, $\mathbf{z}_j^-$, and $\mathbf{w}_v$ are context factors, and b , d , e , h , and g are the biases and global intercepts of each term. The item factor $\mathbf{q}_i$ is shared by the WMF term and both item terms, and the user factor $\mathbf{p}_u$ is shared by the WMF term and the user term. RME therefore uses disliked feedback, but, like CoFactor, it works with co-occurrence between single items or single users rather than group-level patterns.

2.5 Neural and Graph-Based Recommendation

Other feature-free models replace the WMF backbone. NeuMF combines a generalized matrix factorization branch and a multilayer perceptron branch, so it can learn non-linear user-item interactions, and it is trained with sampled negative items [11].

LightGCN learns user and item embeddings by propagating them over the user–item interaction graph with simple weighted sums, and it is trained with a pairwise ranking loss [12]. Both models learn only from the interaction matrix, but they are trained by gradient descent instead of closed-form ALS updates, and they do not add an explicit signal for users with short histories.

2.6 How the Proposed Method Differs

The feature-free methods reviewed above use one or two of clustering, association rule mining, and weighted matrix factorization. CPARMS-WMF combines all three in one framework: clustering provides group-level preference structure shared across the members of each user or item cluster, pairwise association rules provide directional confidence-weighted information, and WMF provides personalized latent-factor ranking. CoFactor and RME are the closest baselines because they also add terms derived from the interaction matrix to the same WMF objective. The main difference is the type of information. CoFactor and RME regularize item and user factors with co-occurrence between single items or single users, while CPARMS builds user–item targets from item–item, user-cluster–item, and item-cluster–item rules. Like RME, CPARMS uses disliked feedback, but it turns this feedback into a separate signal that lowers the scores of candidate items instead of an item co-occurrence term.

CPARMS uses three token families in each signal: user-cluster, item-cluster, and item-level tokens. User-cluster tokens provide a generic signal shared by users in the same cluster, while the other two provide personalized signals based on each user’s own history.

This comparison raises two questions. First, can CPARMS-WMF improve personalized ranking under sparse implicit feedback over the WMF-based baselines, the neural and graph baselines, and the popularity baseline? Second, does this improvement remain for users with very short histories?

3 Proposed Method

CPARMS-WMF is designed to improve Top- K recommendation under sparse implicit feedback. It derives two CPARMS signals, one from liked feedback and one from disliked feedback, through clustering and pairwise association rule mining on the user–item interaction matrix.

Three main components make up the framework: (i) interaction-based clustering of users and items, followed by pairwise association rule mining, run once on the liked view and once on the disliked view; (ii) construction of a sparse user–item signal from the rules of each view; and (iii) integration of both signals into a weighted matrix factorization objective optimized by alternating least squares.

3.1 Notation and Feedback Views

Let $\mathcal{U} = \{1, \dots, U\}$ and $\mathcal{I} = \{1, \dots, I\}$ be the user and item index sets, and let $\mathbf{Y} = [y_{ui}] \in \mathbb{R}_{\geq 0}^{U \times I}$ be the rating matrix of the fit window, with $y_{ui} = 0$ when user u has not rated item i . From $\mathbf{Y}$ the pipeline derives three binary views and keeps them

strictly separate:

$$\underbrace{b_{ui} = \mathbb{I}[y_{ui} > 0]}_{\mathbf{B} \text{ (seen)}}, \quad \underbrace{\ell_{ui} = \mathbb{I}[y_{ui} > \theta]}_{\mathbf{L} \text{ (liked)}}, \quad \underbrace{\delta_{ui} = \mathbb{I}[0 < y_{ui} < \theta]}_{\mathbf{D} \text{ (disliked)}}, \quad \theta = 4. \quad (4)$$

With integer ratings from 1 to 5, a rating of 5 counts as liked, ratings of 1 to 3 count as disliked, and a rating of 4 counts as neither. We use $\theta = 4$ because Amazon ratings are strongly skewed toward high values. With the more common threshold $\theta = 3$, the ratio of liked to disliked ratings in our datasets ranges from 9.9:1 to 40.1:1, which leaves too few disliked ratings for rule mining. With $\theta = 4$, the ratio ranges from 2.8:1 to 12.3:1. A rating of 4 still counts as seen in $\mathbf{B}$.

The liked view $\mathbf{L}$ gives the WMF preference target, the relevance labels for evaluation, and the liked CPARMS signal. The disliked view $\mathbf{D}$ gives only the disliked CPARMS signal. The seen view $\mathbf{B}$ is used to mask candidates: an item is treated as seen after any rating and is not recommended again. For user u , let $\Omega_u^+ = \{i \mid \ell_{ui} = 1\}$ denote the liked history, $\Omega_u^- = \{i \mid \delta_{ui} = 1\}$ the disliked history, $\Omega_u = \{i \mid b_{ui} = 1\} \supseteq \Omega_u^+ \cup \Omega_u^-$ the seen set, and $n_u = |\Omega_u|$ the number of fit-window interactions.

3.2 CPARMS: Clustering and Pairwise Association Rule Construction

Previous studies show that clustering and association rule mining can improve collaborative filtering on sparse data [7, 8]. CPARMS builds on these ideas to construct two additional sparse signals. The same procedure runs twice, once on the liked view and once on the disliked view, with the same hyperparameters. We describe it for a history view $\mathbf{H} \in \{\mathbf{L}, \mathbf{D}\}$ and use the superscripts $+$ and $-$ for the liked and disliked results.

We cluster users and items separately with K-means [16] on $\mathbf{H}$. Each fit uses K-means++ initialization, 10 restarts, and the experiment seed. K-means is fitted only to active rows, which contain at least one entry in $\mathbf{H}$, and the fitted model then assigns every row—including an all-zero row—to its nearest cluster. Excluding all-zero rows from fitting prevents identical inactive rows from forming an uninformative cluster. Because every row is assigned, every user belongs to one user cluster. User clustering operates on the rows of $\mathbf{H}$ and produces K_U clusters; the one-hot matrix $\mathbf{U} \in \{0, 1\}^{U \times K_U}$ records each user’s assignment. Item clustering operates on the rows of $\mathbf{H}^\top$ and produces the one-hot assignment matrix $\mathbf{G} \in \{0, 1\}^{I \times K_I}$. We then construct a binary item-cluster matrix that records whether a user’s history in $\mathbf{H}$ contains any item from each cluster:

$$\mathbf{E} = \mathbb{I}[\mathbf{H}\mathbf{G} > 0] \in \{0, 1\}^{U \times K_I}. \quad (5)$$

Either clustering step can be disabled, which removes the corresponding token family. When $K_U = 1$, all users share the same user-cluster token, which acts as a dataset-wide popularity signal.

For each user, we stack three token families (item-level tokens, user-cluster tokens, and item-cluster tokens) into a single transaction matrix:

$$\mathbf{T} = [\mathbf{H} \mid \mathbf{U} \mid \mathbf{E}] \in \{0, 1\}^{U \times M}, \quad M = I + K_U + K_I. \quad (6)$$

Let $\mathcal{A}$ index all M tokens, and let $\mathcal{R}$ be the item-level tokens, which are the only valid consequents. We use user-cluster and item-cluster tokens only as antecedents, so every rule has an individual item as its consequent.

We mine single-antecedent directed rules from joint token counts. Given the pair-count matrix $\mathbf{C} = \mathbf{T}^\top \mathbf{T}$ and token frequency $n_t = \sum_u T_{ut}$, the metrics for a rule $a \rightarrow b$ are:

$$\text{sup}(a \rightarrow b) = \frac{C_{ab}}{U}, \quad \text{conf}(a \rightarrow b) = \frac{C_{ab}}{n_a}, \quad \text{lift}(a \rightarrow b) = \frac{U C_{ab}}{n_a n_b}. \quad (7)$$

Confidence is the conditional probability that a user linked to antecedent a is also linked to consequent b , and therefore measures the rule’s directional reliability. Lift compares the joint frequency with that expected under independence: $\text{lift} > 1$ indicates a positive association beyond chance, whereas $\text{lift} \approx 1$ indicates a relationship already explained by the consequent’s overall frequency. We consider only directional pairs with $a \neq b$ and $b \in \mathcal{R}$. A rule is selected when it meets the configured minimum support, confidence, and lift. All three thresholds are tuned on validation data (Section 4.5). The rule-selection indicator is

$$F_{ab} = \mathbb{I}[\text{sup}(a \rightarrow b) \geq \sigma_{\min} \wedge \text{conf}(a \rightarrow b) \geq \rho_{\min} \wedge \text{lift}(a \rightarrow b) \geq \ell_{\min}]. \quad (8)$$

Each selected rule keeps its confidence as its weight:

$$W_{ab} = F_{ab} \text{conf}(a \rightarrow b). \quad (9)$$

Thus, retained rules carry weight in proportion to their directional reliability, while support and lift serve only as noise filters through Eq. (8). Collecting the retained rules produces $\mathbf{W} \in \mathbb{R}^{M \times I}$, which encodes the filtered relationships from antecedent tokens to individual-item consequents. On the liked view, a rule $a \rightarrow b$ means that users linked to a often like item b ; on the disliked view, it means that users linked to a often dislike item b .

3.3 CPARMS Signal Generation

Figure 1 summarizes the implemented pipeline. Each view produces two sources of information. Rules triggered by user-cluster tokens in $\mathbf{U}$ give a generic score shared by members of the same cluster. Rules triggered by item-cluster tokens in $\mathbf{E}$ or item-level tokens in $\mathbf{H}$ give personalized scores because both token types depend on the user’s own history in that view.

To form the raw signal, CPARMS applies the rule scores in $\mathbf{W}$ to every token in a user’s transaction row, summing, for each candidate item, the scores of all rules

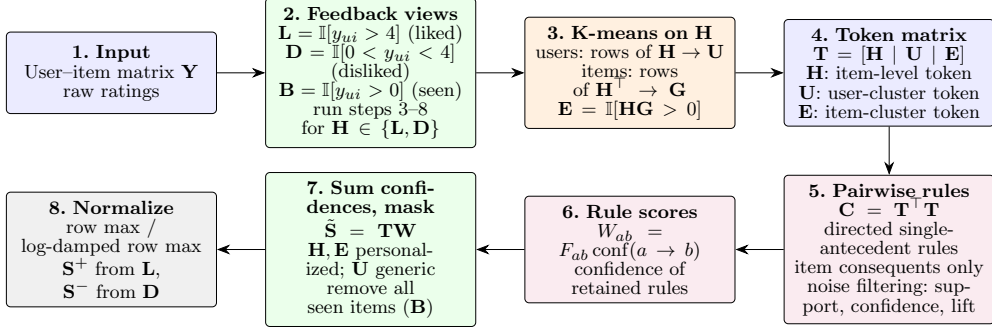


Fig. 1 Overview of the CPARMS signal-generation pipeline, from the raw user-item interaction matrix $\mathbf{Y}$ to the normalized liked and disliked signal matrices $\mathbf{S}^+$ and $\mathbf{S}^-$. Steps 3-8 run once on the liked view $\mathbf{L}$ and once on the disliked view $\mathbf{D}$; the seen view $\mathbf{B}$ enters only as the candidate mask in step 7

triggered by the user's active tokens:

$$\tilde{\mathbf{S}} = \mathbf{T}\mathbf{W} \in \mathbb{R}^{U \times I}. \quad (10)$$

Because the token families occupy separate column blocks of $\mathbf{T}$, this sum divides exactly into the two types of information. We define

$$\mathbf{T}^{(p)} = [\mathbf{H} \mid \mathbf{0}_{U \times K_U} \mid \mathbf{E}], \quad \mathbf{T}^{(g)} = [\mathbf{0}_{U \times I} \mid \mathbf{U} \mid \mathbf{0}_{U \times K_I}]. \quad (11)$$

The raw signal is the equal-weight sum of the personalized and generic signals:

$$\tilde{\mathbf{S}} = \mathbf{T}^{(p)}\mathbf{W} + \mathbf{T}^{(g)}\mathbf{W} = \tilde{\mathbf{S}}^{(p)} + \tilde{\mathbf{S}}^{(g)}. \quad (12)$$

There is no specific blending parameter; the way the two signals combine depends entirely on token activation. Every user activates one user-cluster token, so the generic signal gives a shared cluster-level score vector to each cluster member. In contrast, the number of active personalized antecedents grows with the user's history in the view, so the personalized signal becomes stronger as that history grows. A user with no history in the view activates only the user-cluster token, so Eq. (12) gives this user only the generic signal of that view.

After summing the confidence scores, we set the score of every seen item to zero using $\mathbf{B}$, so the signal applies only to unseen items. Since the seen set includes all liked and disliked items, none of them retains a signal score.

The unseen signal is then normalized using one of two searched modes. Row-wise maximum scaling divides each user row by its own maximum. Log-transformed row-wise scaling first compresses heavy-tailed rule-score sums with $\log(1 + \cdot)$ so that a single dominant item does not compress the remaining scores in the row toward zero.

Let $\tilde{s}_{ui}$ be the unseen raw signal. The two modes are:

$$s_{ui} = \frac{\tilde{s}_{ui}}{\max_j \tilde{s}_{uj}}, \quad \text{or} \quad s_{ui} = \frac{\log(1 + \tilde{s}_{ui})}{\max_j \log(1 + \tilde{s}_{uj})}. \quad (13)$$

Empty rows remain zero, and $0 \leq s_{ui} \leq 1$ under either mode. Running this procedure on $\mathbf{L}$ gives the liked signal $\mathbf{S}^+ = [s_{ui}^+]$, and running it on $\mathbf{D}$ gives the disliked signal $\mathbf{S}^- = [s_{ui}^-]$. Both matrices are sparse and non-negative. We write $\Psi_u^+ = \{i \mid s_{ui}^+ > 0\}$ and $\Psi_u^- = \{i \mid s_{ui}^- > 0\}$ for each user's sets of signal-supported items; by construction, $\Psi_u^+ \cap \Omega_u = \Psi_u^- \cap \Omega_u = \emptyset$. The two sets can overlap, because the same unseen item can be supported by both liked and disliked rules.

3.4 CPARMS-WMF: Signal-Regularized Weighted Matrix Factorization

CPARMS-WMF integrates both signals into a single matrix factorization objective. It adds two CPARMS terms over signal-supported items to implicit-feedback WMF [2], similar to the additional SPPMI terms in CoFactor [9] and RME [10]. We compute interaction scores as the inner product of latent factors:

$$\hat{y}_{ui} = \mathbf{p}_u^\top \mathbf{q}_i, \quad \mathbf{p}_u, \mathbf{q}_i \in \mathbb{R}^K, \quad (14)$$

where K is the latent dimension. Stacking row-wise gives $\mathbf{P} \in \mathbb{R}^{U \times K}$ and $\mathbf{Q} \in \mathbb{R}^{I \times K}$. The preference target is the liked view of Eq. (4), and every pair gets an implicit-feedback confidence that up-weights positives,

$$\pi_{ui} = \ell_{ui}, \quad c_{ui} = 1 + \alpha \pi_{ui}, \quad (15)$$

where $\alpha \geq 0$ sets the extra confidence placed on positive interactions. Disliked and neutral ratings therefore have $\pi_{ui} = 0$ and unit confidence in the WMF term, as in Standard-WMF. The same liked view defines positives for WMF training, for the SPPMI term of CoFactor and the liked SPPMI terms of RME, and for evaluation relevance.

The training objective combines a confidence-weighted WMF term over all user-item pairs, a γ^+ -weighted liked CPARMS term, a γ^- -weighted disliked CPARMS term, and an ℓ_2 penalty,

$$\begin{aligned} \mathcal{L} = & \sum_{u \in \mathcal{U}} \sum_{i \in \mathcal{I}} c_{ui} (\pi_{ui} - \hat{y}_{ui})^2 + \gamma^+ \sum_{u \in \mathcal{U}} \sum_{i \in \Psi_u^+} (s_{ui}^+ - \hat{y}_{ui})^2 \\ & + \gamma^- \sum_{u \in \mathcal{U}} \sum_{i \in \Psi_u^-} (-s_{ui}^- - \hat{y}_{ui})^2 + \lambda (\|\mathbf{P}\|_F^2 + \|\mathbf{Q}\|_F^2). \end{aligned} \quad (16)$$

The first term is the standard implicit WMF loss. It pulls predictions toward 1 on liked items with high confidence $1 + \alpha$ and toward 0 on all remaining pairs with unit confidence. The second term pulls predictions on liked-signal items toward s_{ui}^+ , and

the third term pulls predictions on disliked-signal items toward $-s_{ui}^-$. Every signal-supported item lies outside the seen set, so neither signal competes with an observed rating. For an unseen item supported only by liked rules, the WMF term pulls the prediction toward 0 while the liked term pulls it up, so the item scores higher than other unobserved items. For an unseen item supported only by disliked rules, the disliked term pulls the prediction below 0, so the item scores lower than other unobserved items. When both signals support the same item, both terms are active, and for fixed factors the three terms on that entry are minimized at $\hat{y}_{ui} = (\gamma^+ s_{ui}^+ - \gamma^- s_{ui}^-) / (\gamma^+ + \gamma^-)$. The two CPARMS terms therefore provide additional information: each signal-supported entry receives a nonzero target outside the user’s observed history, whereas the implicit term alone would treat it only as an unobserved zero. Users with short histories gain extra least-squares constraints wherever the rules support unseen candidates. The hyperparameters γ^+ and γ^- set the strength of the two signal terms, and setting either weight to 0 removes that term. The coefficient λ is the ℓ_2 penalty.

3.5 Optimization via Alternating Least Squares

Equation (16) is a weighted least-squares objective. It is quadratic in $\mathbf{P}$ when $\mathbf{Q}$ is fixed and vice versa. For this reason, we optimize CPARMS-WMF with ALS, which allows a closed-form update for each factor row [2]. Let $\mathbf{C}^u = \text{diag}(c_{u1}, \dots, c_{uI})$ be the $I \times I$ confidence diagonal, and let $\boldsymbol{\pi}_u = (\pi_{u1}, \dots, \pi_{uI})^\top$ be the preference vector for user u . Minimizing Eq. (16) over $\mathbf{p}_u$ gives the normal equation:

$$\begin{aligned} & \left(\mathbf{Q}^\top \mathbf{C}^u \mathbf{Q} + \gamma^+ \sum_{i \in \Psi_u^+} \mathbf{q}_i \mathbf{q}_i^\top + \gamma^- \sum_{i \in \Psi_u^-} \mathbf{q}_i \mathbf{q}_i^\top + \lambda \mathbf{I}_K \right) \mathbf{p}_u \\ &= \mathbf{Q}^\top \mathbf{C}^u \boldsymbol{\pi}_u + \gamma^+ \sum_{i \in \Psi_u^+} s_{ui}^+ \mathbf{q}_i - \gamma^- \sum_{i \in \Psi_u^-} s_{ui}^- \mathbf{q}_i. \end{aligned} \quad (17)$$

Following the standard implicit-feedback decomposition, we expand the confidence-weighted terms around the unit-confidence baseline. As a result, we only need to sum the contributions from liked and signal-supported items for each user:

$$\begin{aligned} \mathbf{A}_u &= \mathbf{Q}^\top \mathbf{Q} + \lambda \mathbf{I}_K + \alpha \sum_{i \in \Omega_u^+} \mathbf{q}_i \mathbf{q}_i^\top + \gamma^+ \sum_{i \in \Psi_u^+} \mathbf{q}_i \mathbf{q}_i^\top + \gamma^- \sum_{i \in \Psi_u^-} \mathbf{q}_i \mathbf{q}_i^\top, \\ \mathbf{d}_u &= (1 + \alpha) \sum_{i \in \Omega_u^+} \mathbf{q}_i + \gamma^+ \sum_{i \in \Psi_u^+} s_{ui}^+ \mathbf{q}_i - \gamma^- \sum_{i \in \Psi_u^-} s_{ui}^- \mathbf{q}_i, \end{aligned} \quad (18)$$

where $\mathbf{A}_u \in \mathbb{R}^{K \times K}$ is the user-specific system matrix and $\mathbf{d}_u \in \mathbb{R}^K$ its right-hand side. The user factor follows from solving

$$\mathbf{A}_u \mathbf{p}_u = \mathbf{d}_u. \quad (19)$$

We precompute the shared term $\mathbf{Q}^\top \mathbf{Q}$ once per sweep, so each user update costs only the rank-one contributions of that user’s liked and signal-supported items. Item-factor

updates follow the same procedure with the roles of users and items swapped. We initialize both factor matrices with small-variance random values drawn from a fixed seed, and the algorithm then alternates user and item sweeps for a fixed number of iterations. At inference, we mask all seen items and rank the remaining items for each user by $\hat{y}_{ui} = \mathbf{p}_u^\top \mathbf{q}_i$.

4 Experimental Setup

The experiments use five datasets, a global temporal split, three baselines, and NDCG-based model selection and evaluation.

4.1 Datasets

We use five public benchmark datasets from the 2018 Amazon review corpus [17]: Luxury Beauty, Industrial and Scientific, Prime Pantry, Digital Music, and Musical Instruments. The released 5-core datasets provide the input, and we use only the interaction fields. We then de-duplicate all datasets, keeping the earliest review or, when several reviews share the earliest timestamp, the one with the highest rating. Table 1 reports the resulting statistics.

Table 1 Statistics of the Amazon benchmark datasets after de-duplication

Dataset	#Users	#Items	#Interactions	Int./User	Avg. Rating	Sparsity (%)
Luxury Beauty	3,819	1,581	28,074	7.35	4.24	99.54
Industrial & Scientific	11,041	5,334	72,025	6.52	4.52	99.88
Pantry	14,180	4,970	131,114	9.25	4.54	99.81
Music	16,566	11,797	145,292	8.77	4.70	99.93
Instruments	27,530	10,620	219,156	7.96	4.47	99.93

The published datasets are 5-core based on raw review counts, but our pipeline removes repeated reviews of the same item by the same user, and this step does not preserve the 5-core property. After de-duplication, many users have fewer than five distinct items; in Luxury Beauty, for example, 1,188 of 3,819 users do. For this reason, we describe these datasets as derived from the 5-core releases rather than strictly 5-core, and they are sparser than the 5-core label suggests. Sparsity exceeds 99.5% in every domain. This high sparsity makes the datasets difficult for feature-free methods.

4.2 Data Splitting Protocol

We use a global temporal split to provide a more realistic evaluation setting and avoid training on future interactions [18]. The procedure sorts all events from oldest to newest, using stable input-row order to break timestamp ties. Row position defines an approximate 70/10/20 training, validation, and test split. Each boundary is then moved to the nearest change in timestamp so that events recorded at the same time remain in one partition.

The fit window alone defines which users and items the model can use:

- Tuning stage: fit on training, evaluate on validation.
- Final stage: fit on training-plus-validation, evaluate on test.

Users and items that appear only in a future window never enter the fit matrix as empty rows or columns, so they cannot affect training or appear as candidate items. In particular, they cannot change the U denominator of rule support and lift in Eq. (7). We drop rather than score evaluation events involving unknown users or items. Hyperparameter tuning uses the training and validation sets, after which we retrain the selected configuration on the combined training-plus-validation interactions before the held-out test evaluation.

We divide evaluation users into three positive interaction groups according to the size of their positive history n_u : 1, 2, and 3^+ interactions. An evaluated user must have $n_u \geq 1$ and at least one known positive target in the evaluation window after unknown users and items have been removed. Users with $n_u = 0$ are excluded because an all-zero positive feedback row gives both WMF baselines a zero user factor when solved with ALS. Scoring CPARMS’s cluster-level information against baselines with no user-specific estimate would answer a different question; we leave a fair evaluation of that case for future work.

Under this protocol, the test evaluation covers 1,019 users in Luxury Beauty, 3,904 in Industrial & Scientific, 2,722 in Pantry, 4,080 in Music, and 8,541 in Instruments. Among these users, 110, 456, 343, 462, and 1,017, respectively, have only one positive interaction in the fit window. These evaluated users represent 26.7%, 35.4%, 19.2%, 24.6%, and 31.0% of the total users in each dataset, respectively. Most evaluated users have at least three positive interactions and therefore belong to the 3^+ group.

4.3 Baseline Models

We compare CPARMS-WMF with three feature-free baselines. Every model uses the same temporal splits, relevance threshold $\theta = 3$, metric for model selection, and final test protocol.

MostPop ranks candidate items by global popularity, defined as the number of users who positively interacted with each item in the fit window, and applies the same ranking to every user [13]. MostPop is non-personalized, deterministic, and has no hyperparameters. As a result, its scores stay the same across seeds, and its standard deviation is exactly zero. We include it as a reference point to test whether personalization improves accuracy beyond popularity.

Standard-WMF is the implicit-feedback WMF model of Hu et al. [2]. It uses binary preference $\pi_{ui} = \ell_{ui}$ and confidence $c_{ui} = 1 + \alpha\pi_{ui}$ and learns user and item factors through ℓ_2 regularization and ALS.

CoFactor implements the model in Section 2. It augments the WMF objective with an item-item SPPMI term over the matrix in Eq. (2). The model uses unit weights on nonzero SPPMI entries, separate context factors, item and context biases, and ALS updates for user, item, context, and bias parameters [9].

The search uses CoFactor’s original parameters. It samples the relative WMF scale ℓ and derives $\gamma_{co} = 1/\ell$ and $\lambda_{ctx} = \gamma_{co}\lambda$ instead of treating the co-occurrence

weight and context regularizer as independent parameters. Both search grids come directly from the original paper: $\ell \in \{0.01, 0.05, 0.1, 0.5, 1, 5, 10\}$ and the number of negative samples $k \in \{1, 2, 5, 10, 50\}$ [9]. Sampling these quantities independently could assign negligible weight to the co-occurrence term, effectively reducing CoFactor toward Standard-WMF. We therefore preserve the coupling used in the original parameterization.

4.4 Evaluation Metrics

We measure recommendation performance with NDCG at cutoffs 10, 20, 50, 100, and 200, following standard offline evaluation practice [14]. Relevance is binary: a held-out interaction counts as relevant once its rating exceeds $\theta = 3$. For user u , let $\text{rel}_{u,r} = 1$ when the item ranked at position r is relevant and 0 otherwise, and let IDCG@K_u denote the ideal discounted cumulative gain, that is, the maximum discounted cumulative gain attainable for that user’s known relevant test items at cutoff K . Then

$$\text{NDCG@K}_u = \frac{\sum_{r=1}^K \frac{\text{rel}_{u,r}}{\log_2(r+1)}}{\text{IDCG@K}_u}. \quad (20)$$

For each user, we remove all seen items ($\mathbf{B}$, not only liked items) from the candidate ranking and break score ties by item index. Within each positive interaction group, scores are averaged over the users who meet the criteria in Section 4.2. We report the mean and sample standard deviation over seeds 42–46; all NDCG values are percentages. NDCG@10 is the primary selection metric, while larger cutoffs measure ranking quality over broader candidate lists.

4.5 Hyperparameter Search

We choose model hyperparameters by randomized search [19]. For each seed, we sample 30 configurations per model once and reuse them across all datasets. Every tuned model has a separate reproducible random stream and independently draws all of its hyperparameters, including factorization parameters (K , λ , ALS sweeps, and α) and any model-specific parameters. Models use the same candidate values where a hyperparameter has the same meaning, but values are not forced to match within a trial. The experiment seed controls both factor initialization and K-means. For each dataset–seed pair, we keep each model’s configuration with the best validation NDCG@10, then retrain the model on the combined training-plus-validation set before testing.

Table 2 summarizes the search space. The CPARMS rule thresholds include zero and small positive values because rule-metric scales vary greatly across antecedent families under extreme sparsity. We keep the two auxiliary-term weights separate: CPARMS’s γ weights a user–item signal on a $[0, 1]$ scale, while CoFactor’s γ_{co} weights an SPPMI target whose scale depends on co-occurrence statistics.

Table 2 Hyperparameter search space and descriptions. The search samples thirty configurations per seed and evaluates them on every dataset

Hyperparameter	Search Space	Applies to	Description
Latent dimension K	{10, 20, 30, 40}	all WMF	latent factors
Regularization λ	{0.0001, 0.001, 0.01, 0.1}	all WMF	ℓ_2 penalty
ALS sweeps	{15, 20, 25, 30}	all WMF	ALS iterations
Confidence α	{10, 20, 40, 80}	all WMF	confidence on positives
Signal weight γ	{0.5, 1.0, 2.0, 4.0}	CPARMS	weight of the signal term
Signal normalization	{row_max, log_row_max}	CPARMS	signal scaling mode
User clusters K_U	{None, 1, 2, 3, 5}	CPARMS	user clusters; None disables clustering
Item clusters K_I	{None, 1, 2, 3, 5}	CPARMS	item clusters; None disables clustering
Min. support $\sigma_{\min}$	{0, 0.00005, 0.0001, 0.0003}	CPARMS	support noise filter
Min. confidence $\rho_{\min}$	{0, 0.0005, 0.001, 0.002}	CPARMS	confidence noise filter
Min. lift $\ell_{\min}$	{0, 0.2, 0.5, 0.8}	CPARMS	lift noise filter
Relative WMF scale ℓ	{0.01, 0.05, 0.1, 0.5, 1, 5, 10}	CoFactor	sets $\gamma_{co} = 1/\ell$
SPPMI shift k	{1, 2, 5, 10, 50}	CoFactor	negative samples
Context regularization	derived: $\lambda_{ctx} = \gamma_{co}\lambda$	CoFactor	not searched (derived)

5 Results

We first report overall NDCG and then analyze performance across positive interaction groups. All NDCG values are percentages reported as means over seeds 42–46, with sample standard deviations where applicable. The best values appear in bold; second-best values are underlined. Pairwise seed wins also show whether the advantage appears in many runs or only a few.

5.1 Overall Improvement in Personalized Ranking

The relative improvement of CPARMS-WMF over baseline b is

$$\Delta_b = \frac{\text{NDCG}_{\text{CPARMS-WMF}} - \text{NDCG}_b}{\text{NDCG}_b} \times 100. \quad (21)$$

All Δ_b values are computed from unrounded scores, so they may differ slightly from values recomputed using the rounded means in the tables.

5.1.1 Performance against the WMF baselines

At NDCG@10, CPARMS-WMF wins 73 of the 75 dataset–seed comparisons against the three baselines (Table 3). Against the two WMF baselines alone, it wins 24/25 comparisons each, has the best overall mean on all five datasets, and improves over Standard-WMF by +6.4% (Music), +7.3% (Pantry), +8.9% (Luxury Beauty), +32.7% (Instruments), and +58.4% (Industrial & Scientific). The gains over CoFactor are +5.3% (Music), +7.4% (Luxury Beauty), +10.6% (Pantry), +27.6% (Instruments), and +55.8% (Industrial & Scientific). Its only losses occur on Pantry seed 45 against both WMF baselines. Table 4 provides the corresponding breakdown by positive interaction group. The advantage is not limited to the selection cutoff: Table 5 shows that CPARMS-WMF has the best overall mean at every cutoff from 10 to 200 on all five datasets.

5.1.2 Performance against popularity

MostPop remains an important reference because it beats both WMF baselines overall on Industrial & Scientific and Instruments. Even so, CPARMS-WMF beats MostPop in all 25 overall dataset–seed runs and has a higher mean on every dataset. Its improvement ranges from +12.2% on Industrial & Scientific to +1724.4% on Luxury Beauty. MostPop’s near-zero NDCG@10 (0.25%) causes the very large percentage on Luxury Beauty, where the popularity baseline is weakest. When we weight the five dataset means equally, CPARMS-WMF reaches 2.79%, compared with 2.32% for Standard-WMF, 2.35% for CoFactor, and 1.36% for MostPop.

Table 3 Pairwise NDCG@10 wins by CPARMS-WMF across the 25 dataset–seed runs. Each cell reports wins out of 25 using unrounded scores. The one-interaction row covers users with exactly one positive interaction

Positive interaction group	vs. MostPop	vs. Standard-WMF	vs. CoFactor
All users	25	24	24
1 interaction	13	21	19
2 interactions	18	22	23
3+ interactions	25	23	24

5.2 Positive Interaction Groups

Table 4 divides users by the number of positive interactions. For users with one positive interaction, CPARMS-WMF beats Standard-WMF in 21/25 runs and CoFactor in 19/25. Its largest mean gains over Standard-WMF occur on Industrial & Scientific (+142.6%) and Instruments (+68.2%). The gains are not uniform, however. On Luxury Beauty, the mean for users with one positive interaction is lower than both WMF baselines (9.24 versus 9.53 and 9.79), with only 2/5 seed wins against each; on Music, it is lower than CoFactor. Results against MostPop are mixed: CPARMS-WMF wins 13/25 runs for users with one positive interaction and has the best mean for these users only on Pantry. The evidence therefore supports an improvement over the two WMF baselines in most runs for users with one positive interaction, but not on every dataset and not consistently over popularity.

At two interactions, CPARMS-WMF wins 22/25 against Standard-WMF, 23/25 against CoFactor, and 18/25 against MostPop, and has the best mean on three datasets; the two remaining best means go to MostPop on Industrial & Scientific and CoFactor on Pantry. At three or more interactions, it has the best mean on all five datasets and wins 25/25 against MostPop, 23/25 against Standard-WMF, and 24/25 against CoFactor. This pattern fits the method design: item-level and liked-item-cluster antecedents provide more user-specific information as n_u grows.

5.3 Computational Cost

Table 6 reports training time measured on a machine with an AMD Ryzen 5 7535U processor and 32 GB of RAM. CPARMS-WMF costs more to train than either WMF baseline: total time ranges from 0.334 minutes on Luxury Beauty to 8.818 minutes on Instruments. This is 3.1–14.2 $\times$ the training time of Standard-WMF and 2.1–6.6 $\times$ that of CoFactor. Most of the extra time comes from denser signal-regularized ALS updates rather than signal construction.

5.4 Hyperparameter Behavior

Because we select a configuration for each dataset–seed pair using validation NDCG@10, the chosen CPARMS-WMF configurations vary. Table 7 lists these configurations across the 25 selections:

1. **Cluster counts vary, and the single-cluster setting is common.** Validation selects $K_U = 1, 2, 3, 5$ in 9, 2, 8, 6 runs, respectively, and $K_I = 1, 2, 3, 5$ in 6, 7, 7, 5 runs, respectively. The search space never disables either token family, so the presence of both families is part of the design rather than a finding. The common choice $K_U = 1$ is a special case: every user shares one cluster token, so its rules become a single dataset-wide score vector proportional to the number of users who positively interacted with each item (Eq. (7)). This vector is a popularity-like prior rather than group-specific information. The setting appears in all five Industrial & Scientific runs, where popularity is strongest, while the other datasets mostly use $K_U > 1$.
2. **Rule filtering varies across a broad grid.** Zero thresholds appear 9, 7, and 8 times for support, confidence, and lift, respectively. The most common positive support value is 0.0001 (10 selections). The confidence thresholds 0.0005 and 0.001 each appear 8 times, while the selected lift thresholds span the full grid, with the upper values 0.5 and 0.8 chosen 13 times in total. All searched thresholds are low in absolute terms; even the largest lift threshold, 0.8, allows rules below independence. The selected values are consistent with validation favoring broader signal coverage over strict rule filtering.
3. **The signal weight varies across its full range.** The selected weights are $\gamma = 0.5, 1, 2, 4$ in 8, 2, 8, 7 runs, respectively. The smallest and largest searched values together account for 15 of 25 selections, so no single weight dominates, and the mean of 2.0 reflects this spread rather than a typical setting. Row-max normalization is chosen 16 times and log-damped row-max 9 times.

5.5 Stability Across Seeds

Seed-to-seed variation of CPARMS-WMF is small, as the sample standard deviations in Table 4 show. Each seed changes both the randomized-search draw and model initialization, so this variation reflects the complete selection-and-training procedure. The largest overall standard deviation is 0.18 on Luxury Beauty, followed by about 0.11 on Pantry and 0.10 on Instruments. The one-interaction group varies more, especially

on Luxury Beauty and Instruments, because each user has a shorter interaction history and the evaluated subgroup is smaller.

Table 4 NDCG@10 by positive interaction group, in percent (mean $\pm$ sample standard deviation over seeds 42–46). Groups are defined by the number of positive interactions; “All” averages over all evaluated users. MostPop is deterministic and hyperparameter-free, so its standard deviation is zero by construction. The best mean per dataset and group is in bold and the second-best is underlined

Dataset	Model	All	1	2	3 ⁺
Luxury Beauty	MostPop	0.25 $\pm$ 0.00	1.09 $\pm$ 0.00	0.00 $\pm$ 0.00	0.18 $\pm$ 0.00
	Standard-WMF	4.27 $\pm$ 0.26	<u>9.53</u> $\pm$ 0.41	5.35 $\pm$ 0.71	3.32 $\pm$ 0.23
	CoFactor	4.33 $\pm$ 0.26	9.79 $\pm$ 0.41	5.31 $\pm$ 0.72	3.37 $\pm$ 0.25
	CPARMS-WMF	4.65 $\pm$ 0.18	9.24 $\pm$ 0.79	5.76 $\pm$ 0.71	3.79 $\pm$ 0.18
Industrial & Scientific	MostPop	2.58 $\pm$ 0.00	6.87 $\pm$ 0.00	4.83 $\pm$ 0.00	1.33 $\pm$ 0.00
	Standard-WMF	1.83 $\pm$ 0.11	2.71 $\pm$ 0.16	1.95 $\pm$ 0.22	1.65 $\pm$ 0.19
	CoFactor	1.86 $\pm$ 0.04	2.78 $\pm$ 0.22	1.98 $\pm$ 0.29	<u>1.68</u> $\pm$ 0.09
	CPARMS-WMF	2.89 $\pm$ 0.05	<u>6.58</u> $\pm$ 0.17	<u>4.36</u> $\pm$ 0.22	1.93 $\pm$ 0.06
Pantry	MostPop	1.44 $\pm$ 0.00	<u>2.11</u> $\pm$ 0.00	1.72 $\pm$ 0.00	1.27 $\pm$ 0.00
	Standard-WMF	<u>1.84</u> $\pm$ 0.07	1.75 $\pm$ 0.28	1.68 $\pm$ 0.09	<u>1.88</u> $\pm$ 0.05
	CoFactor	1.78 $\pm$ 0.17	1.54 $\pm$ 0.10	1.81 $\pm$ 0.33	1.82 $\pm$ 0.17
	CPARMS-WMF	1.97 $\pm$ 0.11	2.23 $\pm$ 0.16	<u>1.75</u> $\pm$ 0.19	1.97 $\pm$ 0.13
Music	MostPop	0.44 $\pm$ 0.00	0.59 $\pm$ 0.00	0.43 $\pm$ 0.00	0.41 $\pm$ 0.00
	Standard-WMF	1.72 $\pm$ 0.05	1.74 $\pm$ 0.12	2.73 $\pm$ 0.24	1.53 $\pm$ 0.05
	CoFactor	<u>1.74</u> $\pm$ 0.05	1.87 $\pm$ 0.21	2.73 $\pm$ 0.12	1.53 $\pm$ 0.05
	CPARMS-WMF	1.83 $\pm$ 0.04	<u>1.76</u> $\pm$ 0.14	2.96 $\pm$ 0.13	1.63 $\pm$ 0.05
Instruments	MostPop	2.11 $\pm$ 0.00	4.58 $\pm$ 0.00	2.93 $\pm$ 0.00	1.56 $\pm$ 0.00
	Standard-WMF	1.95 $\pm$ 0.04	2.29 $\pm$ 0.19	2.14 $\pm$ 0.15	1.86 $\pm$ 0.05
	CoFactor	2.03 $\pm$ 0.07	2.35 $\pm$ 0.21	2.22 $\pm$ 0.09	<u>1.94</u> $\pm$ 0.10
	CPARMS-WMF	2.59 $\pm$ 0.10	<u>3.85</u> $\pm$ 0.48	3.19 $\pm$ 0.12	2.27 $\pm$ 0.08

Table 5 Overall NDCG at multiple ranking cutoffs, in percent (mean $\pm$ sample standard deviation over seeds 42–46). NDCG@10 is the primary model-selection metric. The best mean per dataset and cutoff is in bold and the second-best is underlined

Dataset	Model	NDCG@10	NDCG@20	NDCG@50	NDCG@100	NDCG@200
Luxury Beauty	MostPop	0.25 $\pm$ 0.00	1.43 $\pm$ 0.00	2.94 $\pm$ 0.00	3.49 $\pm$ 0.00	4.42 $\pm$ 0.00
	Standard-WMF	4.27 $\pm$ 0.26	4.96 $\pm$ 0.22	6.08 $\pm$ 0.21	6.88 $\pm$ 0.21	7.87 $\pm$ 0.19
	CoFactor	<u>4.33 $\pm$ 0.26</u>	<u>4.97 $\pm$ 0.26</u>	<u>6.09 $\pm$ 0.19</u>	<u>6.90 $\pm$ 0.15</u>	<u>7.97 $\pm$ 0.15</u>
	CPARMS-WMF	4.65 $\pm$ 0.18	5.43 $\pm$ 0.17	6.58 $\pm$ 0.19	7.46 $\pm$ 0.25	8.53 $\pm$ 0.20
Industrial & Scientific	MostPop	<u>2.58 $\pm$ 0.00</u>	<u>3.01 $\pm$ 0.00</u>	<u>3.81 $\pm$ 0.00</u>	<u>4.54 $\pm$ 0.00</u>	<u>5.40 $\pm$ 0.00</u>
	Standard-WMF	1.83 $\pm$ 0.11	2.35 $\pm$ 0.11	3.20 $\pm$ 0.09	4.01 $\pm$ 0.07	4.97 $\pm$ 0.10
	CoFactor	1.86 $\pm$ 0.04	2.35 $\pm$ 0.06	3.23 $\pm$ 0.14	4.02 $\pm$ 0.14	4.98 $\pm$ 0.19
	CPARMS-WMF	2.89 $\pm$ 0.05	3.47 $\pm$ 0.10	4.40 $\pm$ 0.22	5.20 $\pm$ 0.29	6.17 $\pm$ 0.37
Pantry	MostPop	1.44 $\pm$ 0.00	2.07 $\pm$ 0.00	3.20 $\pm$ 0.00	4.47 $\pm$ 0.00	<u>5.90 $\pm$ 0.00</u>
	Standard-WMF	<u>1.84 $\pm$ 0.07</u>	<u>2.37 $\pm$ 0.07</u>	<u>3.46 $\pm$ 0.08</u>	<u>4.48 $\pm$ 0.14</u>	5.79 $\pm$ 0.19
	CoFactor	1.78 $\pm$ 0.17	2.35 $\pm$ 0.14	3.39 $\pm$ 0.11	4.44 $\pm$ 0.14	5.71 $\pm$ 0.13
	CPARMS-WMF	1.97 $\pm$ 0.11	2.56 $\pm$ 0.09	3.76 $\pm$ 0.12	5.00 $\pm$ 0.21	6.42 $\pm$ 0.36
Music	MostPop	0.44 $\pm$ 0.00	0.64 $\pm$ 0.00	0.98 $\pm$ 0.00	1.51 $\pm$ 0.00	2.00 $\pm$ 0.00
	Standard-WMF	1.72 $\pm$ 0.05	2.19 $\pm$ 0.07	3.06 $\pm$ 0.07	<u>3.98 $\pm$ 0.09</u>	5.09 $\pm$ 0.10
	CoFactor	<u>1.74 $\pm$ 0.05</u>	<u>2.22 $\pm$ 0.09</u>	<u>3.06 $\pm$ 0.07</u>	<u>3.97 $\pm$ 0.09</u>	5.06 $\pm$ 0.10
	CPARMS-WMF	1.83 $\pm$ 0.04	2.31 $\pm$ 0.04	3.20 $\pm$ 0.07	4.19 $\pm$ 0.06	5.32 $\pm$ 0.03
Instruments	MostPop	<u>2.11 $\pm$ 0.00</u>	2.59 $\pm$ 0.00	3.60 $\pm$ 0.00	4.46 $\pm$ 0.00	5.42 $\pm$ 0.00
	Standard-WMF	1.95 $\pm$ 0.04	2.62 $\pm$ 0.06	3.65 $\pm$ 0.08	<u>4.62 $\pm$ 0.10</u>	<u>5.72 $\pm$ 0.14</u>
	CoFactor	2.03 $\pm$ 0.07	<u>2.64 $\pm$ 0.05</u>	<u>3.65 $\pm$ 0.08</u>	4.60 $\pm$ 0.13	5.69 $\pm$ 0.19
	CPARMS-WMF	2.59 $\pm$ 0.10	3.35 $\pm$ 0.10	4.51 $\pm$ 0.13	5.51 $\pm$ 0.18	6.60 $\pm$ 0.27

Table 6 Training time components in minutes (mean $\pm$ sample standard deviation over five seeds). “Signal” is signal-matrix construction; “ALS” is factorization optimization time

Dataset	Model	Signal	ALS training	Total
Luxury Beauty	MostPop	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
	Standard-WMF	0.000 $\pm$ 0.000	0.108 $\pm$ 0.028	0.108 $\pm$ 0.028
	CoFactor	0.000 $\pm$ 0.000	0.162 $\pm$ 0.019	0.162 $\pm$ 0.019
	CPARMS-WMF	0.010 $\pm$ 0.002	0.324 $\pm$ 0.107	0.334 $\pm$ 0.105
Industrial & Scientific	MostPop	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
	Standard-WMF	0.000 $\pm$ 0.000	0.344 $\pm$ 0.142	0.344 $\pm$ 0.142
	CoFactor	0.001 $\pm$ 0.000	0.533 $\pm$ 0.181	0.534 $\pm$ 0.181
	CPARMS-WMF	0.052 $\pm$ 0.017	2.251 $\pm$ 0.976	2.303 $\pm$ 0.991
Pantry	MostPop	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
	Standard-WMF	0.000 $\pm$ 0.000	0.329 $\pm$ 0.114	0.329 $\pm$ 0.114
	CoFactor	0.002 $\pm$ 0.000	0.629 $\pm$ 0.174	0.631 $\pm$ 0.174
	CPARMS-WMF	0.073 $\pm$ 0.029	2.379 $\pm$ 0.806	2.452 $\pm$ 0.832
Music	MostPop	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
	Standard-WMF	0.000 $\pm$ 0.000	0.523 $\pm$ 0.156	0.523 $\pm$ 0.156
	CoFactor	0.003 $\pm$ 0.000	1.145 $\pm$ 0.271	1.148 $\pm$ 0.271
	CPARMS-WMF	0.086 $\pm$ 0.051	4.219 $\pm$ 2.656	4.305 $\pm$ 2.707
Instruments	MostPop	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000
	Standard-WMF	0.000 $\pm$ 0.000	0.619 $\pm$ 0.197	0.619 $\pm$ 0.197
	CoFactor	0.003 $\pm$ 0.000	1.341 $\pm$ 0.279	1.344 $\pm$ 0.279
	CPARMS-WMF	0.217 $\pm$ 0.174	8.602 $\pm$ 6.372	8.818 $\pm$ 6.542

Table 7 Selected CPARMS-WMF hyperparameter configuration for each dataset and seed (best validation NDCG@10). K : latent dimension; sweeps: ALS iterations; α : confidence boost; γ : signal weight; K_U/K_I : user/item clusters; $\sigma_{\min}/\rho_{\min}/\ell_{\min}$: minimum rule support, confidence, and lift; norm: signal normalization (row: row max; log-row: log-damped row max)

Dataset	Seed	K	λ	sweeps	α	γ	K_U	K_I	$\sigma_{\min}$	$\rho_{\min}$	$\ell_{\min}$	norm
Luxury Beauty	42	40	0.1	30	40	0.5	3	3	0	0.001	0.8	row
	43	40	0.1	20	10	0.5	5	2	0	0.0005	0.8	row
	44	40	0.0001	20	80	2	1	2	0	0	0	row
	45	20	0.1	15	80	0.5	3	3	0.0001	0.0005	0.2	row
	46	40	0.0001	15	80	2	5	5	0.0003	0.001	0.8	row
Industrial & Scientific	42	40	0.01	25	20	2	1	1	0.0003	0	0.5	row
	43	40	0.01	20	80	4	1	2	0.0003	0	0.2	log-row
	44	40	0.001	20	20	1	1	1	0.0003	0	0.5	log-row
	45	40	0.1	15	80	4	1	1	0.0001	0.001	0	row
	46	30	0.001	15	20	4	1	1	0.00005	0.0005	0.5	log-row
Pantry	42	40	0.0001	20	80	4	3	5	0.0001	0.002	0.8	row
	43	30	0.0001	20	80	4	5	5	0.0003	0	0	row
	44	20	0.01	30	80	2	1	3	0.0001	0.0005	0	log-row
	45	30	0.0001	15	20	4	3	2	0.0001	0	0.5	log-row
	46	40	0.1	20	20	2	5	3	0.0001	0.001	0.2	log-row
Music	42	40	0.1	30	40	0.5	3	3	0	0.001	0.8	row
	43	40	0.1	25	80	0.5	3	3	0.0001	0.001	0	row
	44	40	0.001	15	80	1	2	1	0.0001	0.001	0.5	row
	45	30	0.001	30	80	0.5	2	5	0.0001	0.001	0.5	log-row
	46	40	0.0001	30	40	0.5	5	5	0	0.0005	0.2	log-row
Instruments	42	30	0.01	15	80	4	1	3	0.0001	0.0005	0	row
	43	40	0.1	20	10	0.5	5	2	0	0.0005	0.8	row
	44	40	0.0001	20	80	2	1	2	0	0	0	row
	45	30	0.1	30	40	2	3	1	0	0.002	0	row
	46	30	0.1	30	20	2	3	2	0	0.0005	0.5	log-row

6 Discussion

CPARMS-WMF has higher mean NDCG@10 than both WMF baselines on all five datasets, with the largest gains on Industrial & Scientific and Instruments. These are also the domains in which popularity is relatively strong, suggesting that the generic component may help when sparse histories make the WMF estimates less reliable.

CoFactor is the closest baseline because both models are feature-free extensions of the same WMF objective. CoFactor slightly exceeds Standard-WMF on four of the five dataset means, whereas CPARMS-WMF exceeds CoFactor on all five means and in 24/25 runs. Under this evaluation, CPARMS-WMF outperforms CoFactor.

Performance for the one-interaction group is less consistent. CPARMS-WMF generally improves on the WMF baselines but does not consistently outperform MostPop, indicating that one positive interaction may provide too little user-specific information in popularity-dominated domains.

The advantage over the WMF baselines grows as the number of positive interactions increases. At two interactions, CPARMS-WMF wins 22/25 comparisons against Standard-WMF and 23/25 against CoFactor. At three or more interactions, it has the best mean on all datasets. However, MostPop beats both WMF baselines overall on Industrial & Scientific and Instruments.

On Industrial & Scientific, where popularity is strongest and both WMF baselines trail MostPop, validation selects $K_U = 1$ in all five runs (Section 5). The generic component then becomes a popularity-like prior, and CPARMS-WMF is the only personalized model to beat MostPop. Validation selects $K_U > 1$ on Music and in four of the five Luxury Beauty runs. As n_u grows, users activate more item and liked-item-cluster tokens, increasing the personalized part of the signal (Section 3). CPARMS-WMF can thus use a popularity-like component in a popularity-dominated domain while retaining cluster- and item-level information elsewhere.

The size and source of the gain vary across datasets. Music has the smallest CPARMS-WMF margin over Standard-WMF (+6.4%), while Industrial & Scientific has the largest (+58.4%). The strongest competitor for users with one positive interaction also varies by domain: CoFactor on Luxury Beauty and Music, while popularity remains difficult to beat for these users on Industrial & Scientific and Instruments. Pantry is the only dataset where CPARMS-WMF has the best mean for users with one positive interaction.

7 Limitations

This study has three main limitations.

1. **Popularity stays competitive at the shortest histories.** CPARMS-WMF improves over Standard-WMF and CoFactor in most comparisons for users with one positive interaction. However, MostPop has the best mean for users with one positive interaction on Industrial & Scientific and Instruments and the best mean for users with two positive interactions on Industrial & Scientific. As a result, the method improves on the WMF baselines but does not always beat a non-personalized baseline.

2. **Computational cost.** CPARMS-WMF is slower than both WMF baselines, especially on the larger datasets, because the additional signal increases the work required for ALS updates.
3. **Limited base-model integration.** The main contribution of this study is the CPARMS signal, which is designed to augment a base recommender; in this work, we integrate it only with WMF. We have not evaluated whether CPARMS can be combined effectively with other architectures, such as graph-based or neural recommender models.

8 Conclusion

We examined whether interaction data alone can improve personalized recommendation under sparse implicit feedback. The proposed **CPARMS-WMF** framework clusters users and items from positive feedback, mines single-antecedent association rules over item, user-cluster, and liked-item-cluster tokens, and incorporates the resulting signal through an additional ALS-optimized term.

Under a global temporal split, CPARMS-WMF beats MostPop, Standard-WMF, and CoFactor in 73 of the 75 dataset–seed comparisons. It improves NDCG@10 over Standard-WMF on all five Amazon datasets (+6.4% to +58.4%), wins 24/25 runs against each WMF baseline and 25/25 against MostPop, and has the best overall mean on all five datasets.

For users with exactly one positive interaction, CPARMS-WMF beats Standard-WMF in 21/25 seed runs and CoFactor in 19/25. The gains are not consistent across datasets: Luxury Beauty is the clearest exception, with both WMF baselines having higher one-interaction means. The results do not show consistent wins over popularity; CPARMS-WMF wins only 13/25 one-interaction runs against MostPop.

Future work could extend CPARMS beyond pairwise rules with single antecedents. The current design keeps rule mining efficient and interpretable, but higher-order antecedents may capture combinations of liked items or item clusters that cannot be expressed by a single token. Comparative experiments with Apriori or FP-Growth should therefore evaluate whether these richer patterns improve ranking accuracy enough to justify their additional mining cost [20]. Another priority is a fair evaluation protocol for users with no positive fit-window history.

Finally, this study has limited base-model integration. Although the CPARMS signal is designed to augment a base recommender, we evaluate it only with WMF. It therefore remains unclear whether CPARMS can be combined effectively with other architectures. Future work should integrate the signal with graph-based and neural recommender models to determine whether its benefits generalize beyond WMF.

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Competing interests. B.A. is employed by KASIKORN Business-Technology Group (KBTG), which supported this work. P.T. and A.T. declare no competing interests.

Ethics approval. Not applicable. This study uses an existing public review dataset and does not involve participant recruitment, intervention, or animal research.

Consent to participate. Not applicable.

Consent for publication. Not applicable.

Data availability. The Amazon review datasets used in this study are publicly available from the McAuley lab data repository at <https://nijianmo.github.io/amazon/> [17]. Section 4 fully describes the experimental configurations, including the hyperparameter search space and the evaluation protocol, to support reproducibility.

Code availability. The implementation of CPARMS-WMF and the evaluation pipeline is publicly available at <https://github.com/peerapat-t/cparms-wmf-recsys>.

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